MUKESH KUMAR S

BIOMEDICAL ENGINEER | MEDICAL DEVICE R&D | HEALTHCARE AI & MEDICAL IMAGING

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Biomedical Instrumentation • Embedded Systems • Medical Imaging • Wearable Health Technology

PROFESSIONAL SUMMARY

Entry-level Biomedical Engineering graduate (B.E., 2026) with project experience spanning biomedical instrumentation, embedded systems, medical imaging, healthcare AI, and MATLAB-based biomechanical simulation. Developed a desktop chest X-ray classification application using PyTorch and MobileNetV2, integrating X-ray validation, image-quality assessment, Grad-CAM visualisation, probability display, and PDF reporting. Familiar with Python, C/C++, MATLAB, PySide6, ESP32, MAX30205, MAX30102, and BLE. Seeking fresher opportunities across biomedical engineering, medical-device R&D, and healthcare AI/research.

TECHNICAL SKILLS

Programming: Python, C, C++, MATLAB
Healthcare AI & Medical Imaging: PyTorch, MobileNetV2, chest X-ray classification, five-class probability visualisation, Grad-CAM
Desktop Application & Reporting: PySide6, image processing, image-quality assessment, PDF report generation
Embedded Systems & Sensors: Arduino, ESP32, firmware, BLE, MAX30205 temperature sensor, MAX30102 pulse oximetry/heart-rate sensor, MEMS microphone
Biomedical Instrumentation: Physiological measurement, medical equipment exposure, calibration concepts
Standards Awareness: ISO 13485, ISO 14971, IEC 60601-1
Tools: MATLAB, Arduino IDE, MS Excel

SELECTED PROJECTS

AI-Assisted Chest X-ray Disease Prediction & Explainability — 2026

Python | PyTorch | MobileNetV2 | Grad-CAM | PySide6

- Developed a desktop application for five-class chest X-ray classification: COVID, Lung Opacity, Normal, Pneumonia, and Tuberculosis.
- Integrated X-ray validation, image-quality assessment, class-probability visualisation, and Grad-CAM visual explanations.
- Implemented automated PDF report generation containing patient-entered details, prediction results, image-quality information, and visual explanations.
- Created a graphical interface for research and educational decision-support workflows; outputs are not a standalone medical diagnosis.

Project links: **Live Demo** | **GitHub Repository**

AI-Based Multi-Modal Neonatal Health Assessment System — 2026

Concept design | MEMS microphone | MAX30205 | MAX30102 | ESP32 | BLE

- Designed a proposed wearable monitoring concept combining baby-cry/audio, temperature, SpO2, and heart-rate acquisition.
- Explored integration of a MEMS microphone, MAX30205, MAX30102, ESP32, and BLE for remote monitoring.

Multilabel Classification of Lung Diseases Using Deep Learning — 2026

Deep learning | Chest X-ray analysis

- Investigated deep-learning-based multilabel analysis of chest X-rays for pneumonia, tuberculosis, and COVID-related abnormalities.

Smart Biomedical Thermometer Using MAX30205 — 2026

Arduino | MAX30205 | Embedded firmware

- Developed an Arduino-based temperature measurement prototype using MAX30205 with temperature acquisition and monitoring firmware.

Simulation of 3D Knee Contact Model Using MATLAB — 2025–2026

MATLAB | Biomechanical simulation

- Simulated knee-joint contact behaviour and biomechanical loading using MATLAB.

EDUCATION

B.E. Biomedical Engineering | E.G.S. Pillay Engineering College, Nagapattinam | 2022–2026 | CGPA: 7.43/10
Higher Secondary – Bio-Maths | Laurel Higher Secondary School | 2021–2022 | 75%

INDUSTRIAL & HOSPITAL EXPOSURE

Guru Multispecialty Hospital, Madurai | May 2025 — Biomedical equipment, medical instruments, and clinical workflow exposure.
Dr. Rao Hospital, Madurai | May 2025 — Medical instruments and biomedical equipment exposure.
ITI, Madurai | 2025 — Industrial and technical exposure.

CERTIFICATIONS & TRAINING

- Signal Processing Onramp — MathWorks
- Advanced Critical Care Equipment Training and Calibration Training — KAV Biomedical Equipment Training and Education
- Clinical Knowledge and Medical Coding in Healthcare System — Theeram Healthcare, Madurai
- Microbial Biotechnology — 3-credit NPTEL/SWAYAM course, Jan–Apr 2025

ACHIEVEMENTS

- Represented the Biomedical Engineering Department at Science Expo, E.G.S. Pillay Engineering College, August 2024.
- Recognized for participation in Science Expo 23.
- Participated in technical sessions on Biomedical Instrument Design and Generative AI.
- Participated in an inter-college quiz organized by Shankar IAS Academy.